

RESEARCH STATEMENT

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As an applied microeconomist, my research examines how the design of educational policies and information markets shapes strategic behavior. I investigate how well-intentioned policies, from school entry-age cutoffs to quality certification systems, can be undermined by the very incentives they create, leading to unintended consequences for student opportunity and market efficiency. This focus is grounded in my professional experience within Chile's higher education system, which provided a first-hand perspective of the strategic interactions between students, universities, and regulators. My dissertation builds this agenda by systematically analyzing strategic behavior at different levels of the education market. It begins with the household-level decision of school entry timing, creating disparities tied to family background; it then shifts to the firm-level behavior of universities gaming quality signals and the subsequent impact on student choice; and finally, it examines the industrial organization of the certification market itself, testing whether competition among accrediting agencies leads to signal inflation. My approach leverages Chile's unique institutional setting and rich administrative data as an ideal scenario to answer these questions, yielding causal insights relevant for policy design worldwide.

Job market paper

My job market paper, "Signaling Quality or Gaming the System? Evidence from College-Major Accreditation," investigates how quality certification functions in markets characterized by asymmetric information. My analysis is centered on the student response. By using rich administrative data and a difference-in-differences framework that exploits the staggered timing of first-time accreditation, I estimate the causal effect of this information shock. I find that students react powerfully as first-time accreditation causes a 10% increase in applications and a 7% increase in enrollment. This demand shock is driven by academically stronger students and, remarkably, the effects are consistent across socioeconomic status, appearing just as strong for students from low-income backgrounds. This initial finding demonstrates that the signal is broadly effective at improving student sorting and access. The heterogeneity of these effects reveals further variation, as we analyze the impact by institutional reputation. We find that the signal's value is amplified for programs within already reputable universities, suggesting that formal certification and informal reputation act as complements. Furthermore, we show that the signal's power is substantially weakened under a mandatory regulatory framework. In this regime, where accreditation is universal, students tend to disregard the binary status and interpret the accreditation's duration as the effective indicator of quality.

The paper then analyzes the nature of the signal students are responding to, focusing on policy trade-offs. Consistent with expectations that the evaluation framework focuses on student outcomes, we find no evidence of anticipatory increases in long-term structural inputs, like faculty or infrastructure. However, a striking finding emerges in more malleable metrics, as we document a sudden and significant increase in on-time graduation rates concentrated precisely before a program's first evaluation, which is not sustained in the post-accreditation period. Critically, the heterogeneity analysis of this finding reveals that this behavioral response is concentrated at middle and bottom-tier institutions. This temporal pattern suggests institutional management timed to the assessment cycle, raising questions about productive efficiency. This highlights a crucial challenge for policymakers, which is designing a quality assurance system that provide clear signals without incentivizing compliance focused on short-term evaluated metrics.

Additional Projects

In my second chapter, "School Starting Age and the College Application Margin: Evidence from Chile," I investigate the long-run consequences of being older or younger than one's classmates. It's a well-known phenomenon that older students in a grade often do better, but my paper asks a simple question: Is this a true developmental advantage, or is it driven by strategic family decisions? By exploiting sharp school-entry cutoff dates in Chile, I find that starting school a year later initially makes students

about 2 percentage points more likely to enroll in college and boosts their high school GPA by 0.08 standard deviations. The crucial finding, however, is that the college enrollment advantage disappears completely once I account for parental income and education. This suggests the “age advantage” is not simply about maturity, but is actually a reflection of socioeconomic status, as better-resourced families are the ones who strategically delay their children’s school entry. This practice has significant welfare implications, suggesting that the ability to manipulate enrollment timing can displace talented, younger students from less-advantaged backgrounds in the competitive process for college admissions.

My third chapter, “Grading the Graders: Competition and Quality Inflation in Certification Markets,” explores what happens when the organizations that provide quality ratings have to compete for business. The project asks whether competition among college accrediting agencies leads them to become more lenient in order to attract universities as clients, a phenomenon known as a “race to leniency.” To answer this, I take advantage of an expansion in the market of accreditation agencies, combined with a policy change that suddenly incorporated incentives for education programs to seek accreditation, creating a natural experiment. My preliminary findings show that this surge in demand led to signal inflation: education programs started receiving longer accreditation periods, suggesting more lenient evaluations. This project goes a step further by using textual analysis of the full accreditation reports, providing a richer way to measure leniency and understand how these rating agencies strategically use language.

Future research agenda

My future research will extend my dissertation work by investigating the welfare consequences of gamed information signals. This leads to a crucial question: do students see through the game? While my job market paper shows accreditation improves the initial student-program match, the long-run effects are unknown. By linking my accreditation data to post-graduation employment records from administrative sources, I will test whether students attracted to programs by lenient or “inflated” accreditation signals experience worse labor market outcomes, thereby measuring the tangible welfare costs of signal distortion.

Second, I aim to deepen my analysis of student choice by investigating the gap between beliefs and reality in educational investment. Public information on graduate earnings is often coarse, masking significant variation across institutions. By contrasting these public signals with realized earnings, I will investigate how accurately student application decisions reflect their likely financial outcomes. Using a discrete choice framework, I will then run counterfactual simulations to estimate how student sorting and market efficiency would change if students had access to more precise information.

Finally, my research agenda will broaden its focus from how universities respond to quality signals to how the market for these signals is structured. This new project investigates the industrial organization of the certification market itself. I plan to model private accreditation agencies as strategic firms, analyzing how they compete and set their “prices” in terms of leniency to gain market share. Methodologically, this project will move beyond simple outcome measures by using textual analysis and machine learning techniques on the full text of accreditation reports. This will allow me to create nuanced measures of leniency and focus on the specific language that constitutes a persuasive quality signal, enhancing our understanding of how information is strategically produced and supplied.